

B.SC.CSIT 4th Semester
Operating System: CSC259
LAB 03

TITLE

FAMILIARIZATION WITH FIRST IN FIRST OUT (FIFO) PROCESS SCHEDULING ALGORITHM

OBJECTIVES

TO PERFORM FIRST COME FIRST SERVE FIFO SCHEDULING ALGORITHM AND FINDING THE AVERAGE WAITING TIME AND TURNAROUND TIME

REQUIRED THEORY

Given n processes with their burst times, the task is to find average waiting time and average turnaround time using FCFS scheduling algorithm.

First in, first out (FIFO), also known as first come, first served (FCFS), is the simplest scheduling algorithm. FIFO simply queues processes in the order that they arrive in the ready queue.

In this, the process that comes first will be executed first and next process starts only after the previous gets fully executed.

Here we are considering that arrival time for all processes is 0.

SOURCE CODE

```
#include<stdio.h>

int main() {

int n,bt[20],wt[20],tat[20],avwt=0,avtat=0,i,j;

printf("Enter total number of processes(maximum 20):");

scanf("%d",&n);

printf("\nEnter Process Burst Timen");

for(i=0;i<n;i++)
```

```

{
printf("P[%d]:",i+1);

scanf("%d",&bt[i]);

}

wt[0]=0;

for(i=1;i<n;i++)

{

wt[i]=0;

for(j=0;j<i;j++)

wt[i]+=bt[j];

}

printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time");

for(i=0;i<n;i++)

{

tat[i]=bt[i]+wt[i];

avwt+=wt[i];

avtat+=tat[i];

printf("nP[%d]tt%dt\tt%dt\t",i+1,bt[i],wt[i],tat[i]);

}

avwt/=i;

avtat/=i;

printf("\nnAverage Waiting Time:%d",avwt);

printf("\nnAverage Turnaround Time:%d",avtat);

return 0;

}

```

OUTPUT

```
ankit@ubuntu:~$ ls
ankit  Documents  examples.desktop  Music  Public  Videos
Desktop Downloads  lab03            Pictures Templates
ankit@ubuntu:~$ cd lab03
ankit@ubuntu:~/lab03$ ls
process3 process3.c
ankit@ubuntu:~/lab03$ gcc -o process process3.c
ankit@ubuntu:~/lab03$ ls
process process3 process3.c
ankit@ubuntu:~/lab03$ ./process

Enter total number of processes(maximum 20):5

Enter Process Burst Time P[1]: 17
P[2]: 14
P[3]: 19
P[4]:
8
P[5]: 7

Process          Burst Time  Waiting Time  Turnaround Time
P[1]              17           0             17
P[2]              14          17             31
P[3]              19          31             50
P[4]               8          50             58
P[5]               7          58             65
Average Waiting time= 31
Average turnaround time = 44ankit@ubuntu:~/lab03$
```

CONCLUSION

HENCE WE STUDIED AND IMPLEMENTED FIRST IN FIRST OUT (FIFO) PROCESS SCHEDULING ALGORITHM IN LAB WHERE WE CALCULATED AVERAGE WAITING TIME AND AVERAGE TURN AROUND TIME OF PROCESSES IN UBUNTU OPERATING SYSTEM